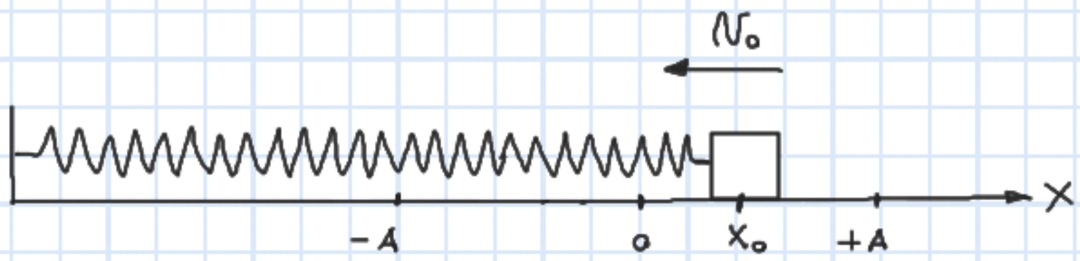


$m = 2\,kg$

$k = 100\,N/m$

$x_0 = 0,1\,m$

$v_0 = -1\,m/s$

$x(t) = A\cos(\omega t + \delta_0)$

$v(t) = -A\omega\,\text{sen}(\omega t + \delta_0)$

$a(t) = -A\omega^2\cos(\omega t + \delta_0)$

a- $\omega = \sqrt{\frac{k}{m}}$

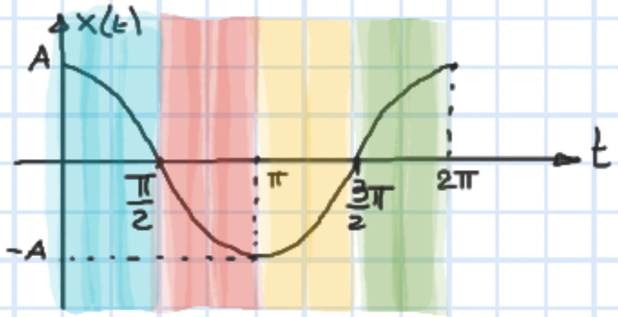
$\omega = 2\pi . f$

$f = \frac{1}{2\pi}\cdot\sqrt{\frac{k}{m}} = \frac{\sqrt{50}}{2\pi}\,Hz$

b- PARA $t=0$

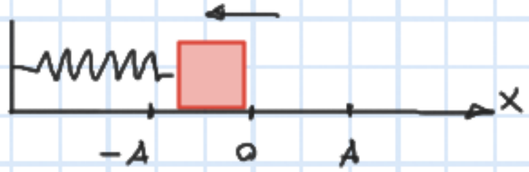
$$\left.\begin{aligned} x(0) &= 0,1\,m = A\cos(\delta_0) \\ v(0) &= -1\,\frac{m}{s} = -A\,\frac{\sqrt{50}}{2\pi}\,\text{sen}(\delta_0) \end{aligned}\right\} \quad tg\,\delta_0 = \frac{20\pi}{\sqrt{50}} \quad \begin{cases} \delta_{01} = 83,58^\circ \\ \delta_{02} = 263,58^\circ \end{cases}$$

$$x(t) = A\cos(\omega t)$$

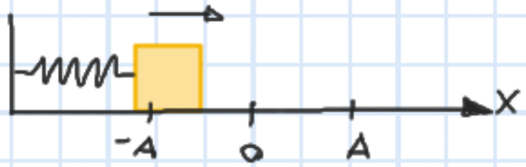


VEAMOS COMO ES EL MOVIMIENTO CON FASE INICIAL CERO

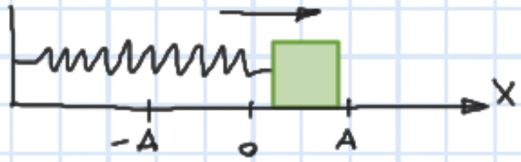
EL TRAMO EN AZUL REPRESENTA EL MOVIMIENTO DEL BLOQUE DESDE LA POSICIÓN $X = A$ A LA DE EQUILIBRIO $X = 0$ $\left(\text{DE } 0 \text{ A } \frac{\pi}{2}\right)$



EL TRAMO EN ROJO REPRESENTA EL MOVIMIENTO DE $X = 0$ A $X = -A$ $\left(\text{DE } \frac{\pi}{2} \text{ A } \pi\right)$



EL TRAMO AMARILLO REPRESENTA EL MOVIMIENTO DE $X = -A$ A $X = 0$ $\left(\text{DE } \pi \text{ A } \frac{3\pi}{2}\right)$



EL TRAMO VERDE REPRESENTA EL MOVIMIENTO DE $X = 0$ A $X = A$ $\left(\text{DE } \frac{3}{2}\pi \text{ A } 2\pi\right)$